

# MAHENDRA PRATAP

Computer Science & AI Engineering Undergraduate · Software Engineering & AI/ML

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## PROFESSIONAL SUMMARY

Computer Science & AI/ML undergraduate with hands-on experience developing autonomous AI agent architectures, local-first web workspaces, and production full-stack systems. Proficient in **Python, TypeScript, React, Next.js**, and modern cloud/database backends. Strong foundation in **Data Structures & Algorithms, Systems Design, and LLM Orchestration** with practical industry exposure to geospatial systems and satellite analytics from **BISAG-N**. Seeking Software Engineering and AI/ML internship opportunities.

## TECHNICAL SKILLS

|                              |                                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| <b>Languages:</b>            | Python, C, C++, TypeScript, JavaScript, SQL                                                                         |
| <b>AI / ML &amp; Agents:</b> | Autonomous Agent Architectures, LLM Orchestration (OpenAI, Gemini), Pydantic, RAG, Prompt Engineering, Scikit-Learn |
| <b>Frontend &amp; UI:</b>    | React, Next.js (App Router), HTML5 Canvas, Tailwind CSS, Framer Motion, Recharts, React Leaflet (GIS)               |
| <b>Backend &amp; Cloud:</b>  | RESTful APIs, SQLite, Firebase (Firestore / Realtime DB), Supabase, IndexedDB, LocalStorage, Vercel                 |
| <b>CS Foundations:</b>       | Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Computer Networks, System Design       |
| <b>Developer Tools:</b>      | Git, GitHub, Linux Shell, VS Code, jsPDF, WebRTC DataChannels, Jupyter Notebook                                     |

## FEATURED PROJECTS

### MEMORA — Autonomous AI Agent & Personal Assistant Engine [github.com/mahendrapratap23/MEMORA ↗](https://github.com/mahendrapratap23/MEMORA)

**Technologies:** Python 3.11+, Pydantic, LLM Orchestration, SQLite, Provider-Agnostic Client, CLI REPL

- Architected an open-source autonomous agent engine in **Python** from scratch featuring a decoupled 6-layer architecture (Interface, Orchestration, LLM, Memory, Tools, and Storage).
- Built a provider-agnostic LLM interface supporting OpenAI, Groq, Ollama, and OpenRouter with strict type validation powered by **Pydantic** and resilient network error recovery.
- Designed long-term persistent session memory using **SQLite** and extensible tool execution pipelines for autonomous multi-step task execution.

### Ghumo AI — Intelligent Travel Planner & Spatial Engine [github.com/mahendrapratap23/ghumo-ai ↗](https://github.com/mahendrapratap23/ghumo-ai)

**Technologies:** Next.js 15, TypeScript, Google Gemini API, React Leaflet (GIS), Recharts, Tailwind CSS, jsPDF

- Developed an AI itinerary planning platform leveraging the **Google Gemini API** with structured prompt engineering to generate personalized multi-day travel schedules and pacing.
- Integrated interactive spatial visualizations with **React Leaflet**, dynamic currency budget charts via **Recharts**, and one-click client-side export via **jsPDF**.
- Designed a responsive UI featuring reusable components, error boundaries, light/dark themes, and automated Vercel deployment.

### Trackly — Local-First Habit & Metric Analytics Suite [github.com/mahendrapratap23/trackly ↗](https://github.com/mahendrapratap23/trackly)

**Technologies:** Next.js 16, React, TypeScript, IndexedDB, LocalStorage, Firebase, Supabase, Recharts

- Implemented a privacy-first, local-first analytics application delivering zero-latency offline metric tracking using **IndexedDB** and **LocalStorage** with optional cloud sync via **Firebase / Supabase**.
- Built analytics dashboards with **5 distinct chart types in Recharts**, automated trend detection, custom date-range filtering, and an interactive HTML5 canvas workspace.

## INDUSTRIAL EXPOSURE & PRACTICAL TRAINING

### Bhaskaracharya National Institute for Space Applications and Geo-informatics (BISAG-N) Gandhinagar, Gujarat

**Technical Focus:** Space Technology, Geographic Information Systems (GIS), Remote Sensing & Geospatial AI

- Completed an on-site industrial technical visit to BISAG-N (MeitY, Govt. of India), gaining practical exposure to national-scale satellite remote sensing, Geo-Informatics, and spatial data systems.
- Explored real-world applications of Space Technology, GIS data pipelines, and the integration of artificial intelligence and machine learning models for satellite image analysis and decision-support systems.

## EDUCATION

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|----------------------------------------------------------------------------------------|--------------------------------------|
| <b>Marwadi University — B.Tech in Computer Science &amp; Engineering (AI &amp; ML)</b> | 2024 – 2028   <b>CGPA: 8.42 / 10</b> |
| <b>Class XII (Senior Secondary) — UP Board (Science &amp; Mathematics)</b>             | <b>88.0%</b>                         |
| <b>Class X (Secondary Education) — UP Board</b>                                        | <b>89.0%</b>                         |

## CERTIFICATIONS & CREDENTIALS

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|----------------------------------------------------------------------|---------------------------------------------------------------------|
| • Oracle Cloud Infrastructure 2025 AI Foundations Associate — Oracle | • IBM Artificial Intelligence Fundamentals — IBM                    |
| • Cisco CCNA & Cloud Networking — Cisco Networking Academy           | • Problem Solving Using Computational Thinking — Univ. of Michigan  |
| • Data Analyst & Generative AI Credentials — LinkedIn Learning       | • Digital Electronics & Hardware Foundations — NPTEL, IIT Kharagpur |